Practice workbook · Calculating price elasticity of demand (PED)

Computing PED as %change in quantity demanded divided by %change in price, including rearranging the formula to find the quantity change.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: ECO-CALC-03

1. Question 1 · part f · 2 marks

2022 JUN · Paper 01 (regional R)

Worked-example retry · Attempt part f only.

1 (a) Which **one** of the following is defined as 'breaking down the production process into smaller tasks'?

(1)

- ☒ **A** Division of labour
- ☒ **B** Innovation
- ☒ **C** Labour intensive
- ☒ **D** Productivity

A firm sells 40 units of a good. Variable costs are $4 per unit. Customers pay $10 per unit.

(b) Which **one** of the following equals $400?

(1)

- ☒ **A** Price
- ☒ **B** Total costs
- ☒ **C** Profit
- ☒ **D** Total revenue

OLevels.COM SCHOOL

(c) What is meant by the term inferior good?

(2)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

1. Question 1 · part f · 2 marks

2022 JUN · Paper 01 (regional R)

Worked-example retry · Attempt part f only.

(d) State **one** reason why a demand curve will shift to the left.

(1)

(e) Define the term production.

(1)

Inny provides a window cleaning service to homeowners in his local area. He recently increased prices by 1.6% and quantity demanded fell by 2.1%.

(f) Calculate, to two decimal places, the **price elasticity of demand (PED)** for Inny's window cleaning service. You are advised to show your working.

(2)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Olevels.com SCHOOL

1. Question 1 · part f · 2 marks

2022 JUN · Paper 01 (regional R)

Worked-example retry · Attempt part f only.

(g) Using the diagram below, draw the likely effects on the market for crops after farmland is damaged by flooding. Label the new curve, the new equilibrium price and the new equilibrium quantity.

(3)

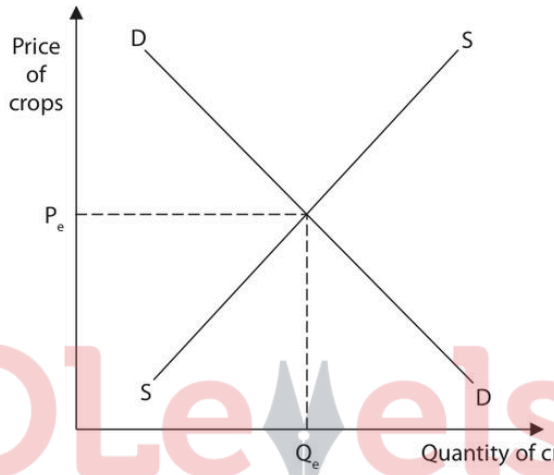


Figure 1

Tertiary sector employment has increased by 13.1% in China during the last 10 years. However, there have been few changes in the level of employment in the secondary sector.

(h) Explain **one** possible reason for this trend.

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

P 6 6 3 2 8 A 0 4 2 4

1. Question 1 · part f · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 2 · part c · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c only.

2 (a) Which **one** of the following is part of the economic problem of scarcity?

(1)

- ☒ **A** When to produce
- ☒ **B** What to produce
- ☒ **C** Whether to produce
- ☒ **D** Where to produce

A country is able to produce agricultural and non-agricultural goods. Its production possibility curve is shown in Figure 3.

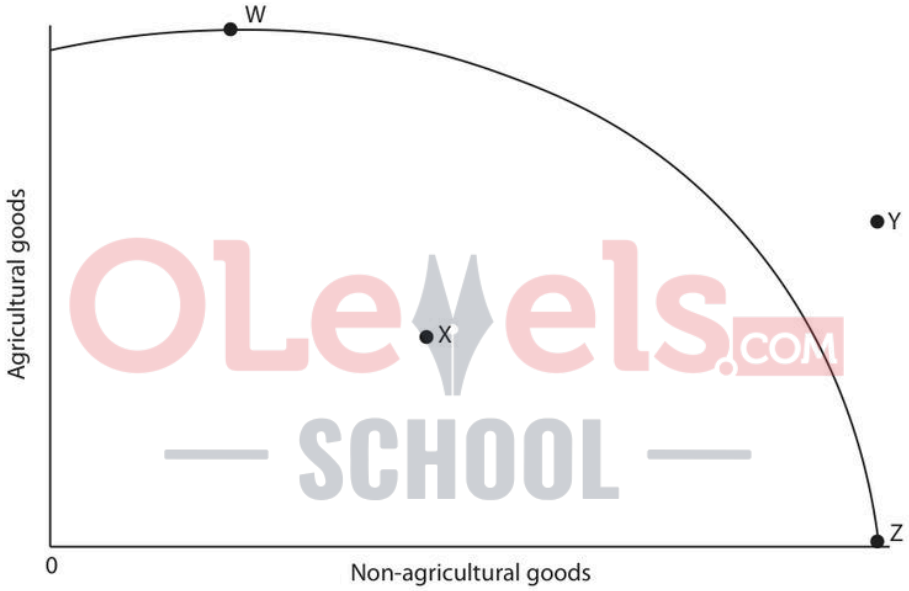


Figure 3

(b) Which **one** of the following points is **not** achievable?

(1)

- ☒ **A** W
- ☒ **B** X
- ☒ **C** Y
- ☒ **D** Z

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 6 0 7 4 9 A 0 6 2 0

2. Question 2 · part c · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

(c) Calculate the price elasticity of demand for a product when price increases by 15% and quantity demanded falls by 12%. You are advised to show your working.

(2)

(d) Define the term innovation.

(1)

(e) State **one** factor that will cause a demand curve to shift to the left.

(1)

OLevels.com SCHOOL

2. Question 2 · part c · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 2 · part c · 2 marks

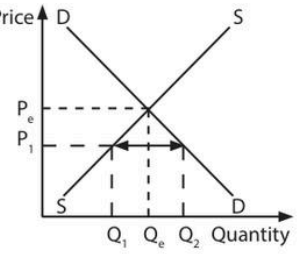
2020 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

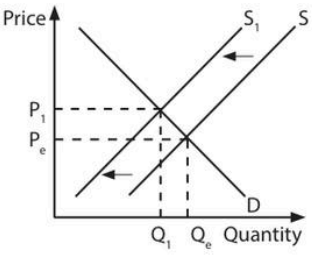
2 (a) Which **one** of the following diagrams shows excess supply?

(1)

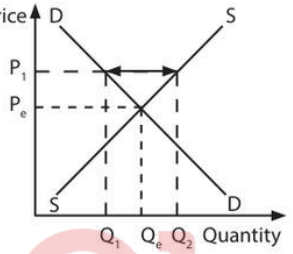
☒ **A**



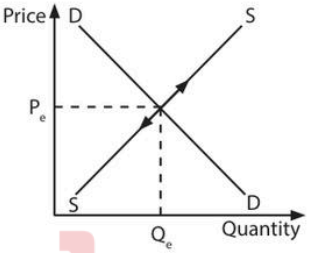
☒ **B**



☒ **C**



☒ **D**



OLevels.com SCHOOL

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

8

P 5 8 2 7 6 A 0 8 2 8

3. Question 2 · part c · 2 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

(b) Which **one** of the following is most likely to create a barrier to entry?

(1)

- ☒ **A** Government provides subsidies for new firms
- ☒ **B** Increasing personal incomes in the economy
- ☒ **C** Low start-up costs
- ☒ **D** Use of expensive technology

(c) Calculate the **price elasticity of demand (PED)** for a product if price falls by 10% and quantity demanded increases by 19%. You are advised to show your working.

(2)

(d) State the formula for social benefits.

(1)

(e) Define the term consumer.

(1)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

3. Question 2 · part c · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 1 · part f · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part f only.

1 (a) Which **one** of the following does a point on a production possibility curve (PPC) represent?

(1)

- ☒ **A** Where capital goods should be produced
- ☒ **B** How the production of all goods can be increased
- ☒ **C** Government revenue from production
- ☒ **D** A given amount of resources being fully employed

Andreas is a plumber. Unlike his competitors, he is prepared to work at the weekend. However, he charges customers 50% more for his services at the weekend.

(b) Which **one** of the following describes this economic assumption?

(1)

- ☒ **A** Businesses aim to maximise their profit
- ☒ **B** Producers may complete charitable work
- ☒ **C** Consumers sometimes copy others' behaviour
- ☒ **D** Governments try to increase the number of days worked

(c) What is meant by the term economies of scale?

(2)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

4. Question 1 · part f · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part f only.

(d) State **one** example of a factor of production that can be classified as land.

(1)

(e) Define the term monopoly.

(1)

Alfie provides a dog-walking service to dog owners in his local area. After a successful first year he increased prices by 1.5% and quantity demanded fell by 1.1%.

(f) Calculate, to two decimal places, the **price elasticity of demand (PED)** for Alfie's dog-walking service. You are advised to show your working.

(2)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Olevels from SCHOOL

4. Question 1 · part f · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part f only.

(g) Using the diagram below, draw the likely effects on the market for cars of the introduction of more effective technology in production. Label the new curve, the new equilibrium price and the new equilibrium quantity.

(3)

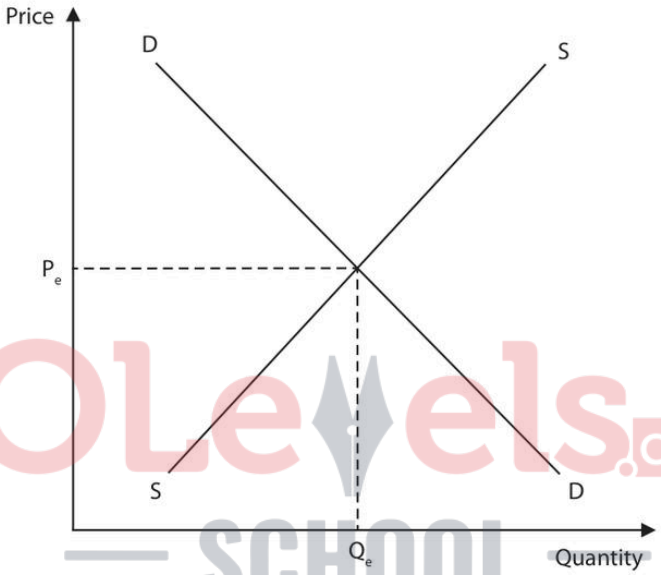


Figure 1

In 2015, the school-leaving age in England was raised from 16 to 18.

(h) Explain **one** possible effect on the supply of labour of the school-leaving age being raised.

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

P 6 5 8 9 6 A 0 4 2 4

4. Question 1 · part f · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 1 · part f · 2 marks

2023 JAN · Paper 01

Further practice · Attempt part f only.

1 (a) What is the name given to a firm that manufactures cars?

(1)

- ☒ **A** A primary sector firm
- ☒ **B** A secondary sector firm
- ☒ **C** A tertiary sector firm
- ☒ **D** A service sector firm

A firm that employs 540 workers has a total monthly output of 27,000 units.

(b) What is the monthly labour productivity of the firm?

(1)

- ☒ **A** 0.02 units
- ☒ **B** 50 units
- ☒ **C** 27,540 units
- ☒ **D** 14,580,000 units

(c) What is meant by the term division of labour?

(2)

OLevels.COM SCHOOL

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

5. Question 1 · part f · 2 marks

2023 JAN · Paper 01

Further practice · Attempt part f only.

(d) State **one** factor of production.

(1)

(e) Define the term public sector.

(1)

Yusef paints portraits of tourists who visit his home town. He recently reduced the price he charges by 2.5% and quantity demanded increased by 3.2%.

(f) Calculate, to two decimal places, the **price elasticity of demand (PED)** for Yusef's portraits. You are advised to show your working.

(2)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Olevels.com — SCHOOL —

5. Question 1 · part f · 2 marks

2023 JAN · Paper 01

Further practice · Attempt part f only.

Figure 1 shows the market for trainers.

(g) Using the diagram below, draw the likely effects on the market after a popular celebrity was seen wearing trainers. Label the new curve, the new equilibrium price and the new equilibrium quantity.

(3)

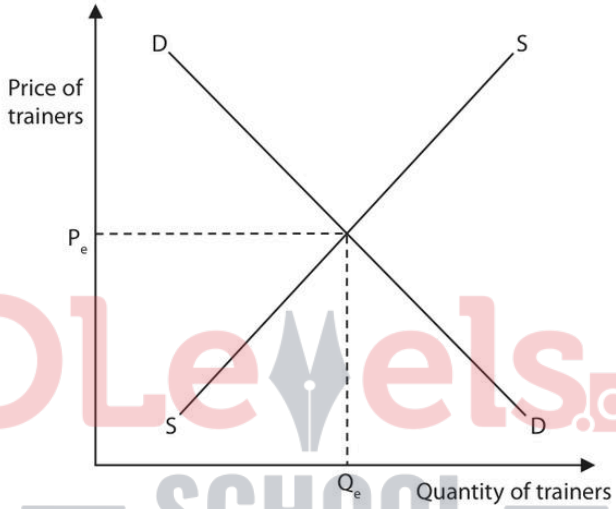


Figure 1

Spice Cottage is a new food producing business. Its prices are lower than market research suggests consumers would be willing to pay.

(h) Explain **one** reason why Spice Cottage may not aim to maximise its profit.

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

4

P 7 1 5 5 5 A 0 4 2 4

5. Question 1 · part f · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 2 · part d · 2 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

2 (a) Which **one** of the following is the formula to calculate price elasticity of supply (PES)? **(1)**

- ☒ **A** $\frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$
- ☒ **B** $\frac{\text{Percentage change in price}}{\text{Percentage change in quantity demanded}}$
- ☒ **C** $\frac{\text{Percentage change in quantity supplied}}{\text{Percentage change in price}}$
- ☒ **D** $\frac{\text{Percentage change in price}}{\text{Percentage change in quantity supplied}}$

A court in China ordered a firm to pay 10 000 Yuan because it polluted a local river.

(b) This is an example of which **one** of the following? **(1)**

- ☒ **A** A fine
- ☒ **B** A permit
- ☒ **C** A subsidy
- ☒ **D** A tax

(c) State **one** example of an external benefit. **(1)**

OLevels.COM SCHOOL

P 7 4 0 5 7 A 0 7 2 4

6. Question 2 · part d · 2 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

Sathvinder provides a cleaning service to homeowners in a small town. She recently decreased prices by 1.2% and quantity demanded increased by 1.9%.

(d) Calculate, to two decimal places, the **price elasticity of demand (PED)** for Sathvinder's cleaning service. You are advised to show your working.

(2)

(e) Define the term entrepreneur.

(1)

(f) Describe **one** reason why water is a factor of production.

(2)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

8

P 7 4 0 5 7 A 0 8 2 4

6. Question 2 · part d · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 2 · part c · 2 marks

2024 JUN · Paper 01

Further practice · Attempt part c only.

DO NOT WRITE IN THIS AREA

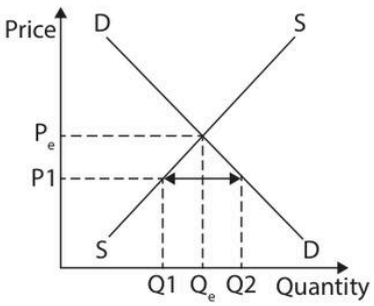
DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

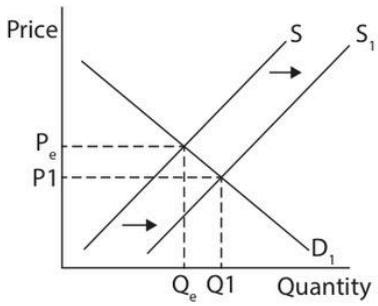
2 (a) Which **one** of the following diagrams shows excess supply?

(1)

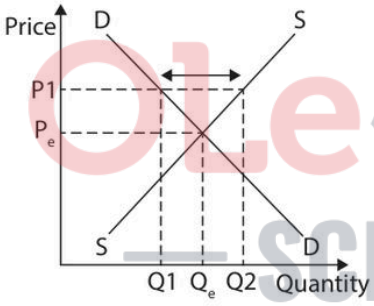
☒ **A**



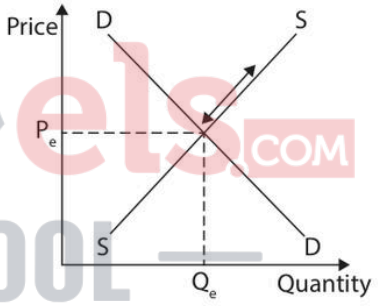
☒ **B**



☒ **C**



☒ **D**



(b) Which **one** of the following is a feature of a monopoly?

(1)

☒ **A** Consumers have a choice of firms

☒ **B** There are many firms competing in the market

☒ **C** The firm is a price-maker

☒ **D** There are low barriers to entry

Olevels.COM SCHOOL

7. Question 2 · part c · 2 marks

2024 JUN · Paper 01

Further practice · Attempt part c only.

(c) Calculate the **percentage change in quantity demanded** for a product if the price elasticity of demand (PED) is –1.9 and the price falls by 10%. You are advised to show your working.

(2)

(d) What is meant by the term social benefits?

(2)

(e) Describe **one** feature of an oligopoly.

(2)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

8

P 7 2 9 6 7 A 0 8 2 4

7. Question 2 · part c · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 1 · part f · 2 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part f only.

1 (a) Which **one** of the following can represent a point on a production possibility curve (PPC)?

(1)

- ☒ **A** Where consumer goods should be produced
- ☒ **B** How the production of all goods can be increased
- ☒ **C** Total revenue from production
- ☒ **D** Production which is obtainable

On one day a month, a firm charges each customer a price of $10 for providing one hour of gardening services. The cost to the firm for providing the service to each customer is $15.

(b) Which **one** of the following describes why the firm might do this?

(1)

- ☒ **A** Producers aim to maximise their profit
- ☒ **B** Producers may complete charitable work
- ☒ **C** Consumers have habits that are hard to give up
- ☒ **D** Consumers sometimes copy others' behaviour

OLevels.COM SCHOOL

(c) What is meant by the term economies of scale?

(2)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

8. Question 1 · part f · 2 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part f only.

DO NOT WRITE IN THIS AREA

(d) State **one** example of an item that can be classified as capital.

(1)

(e) Define the term price-maker.

(1)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(f) Calculate, to two decimal places, the **price elasticity of demand (PED)** for a good if the price decreases by 4.7% and the quantity demanded increases by 1.8%. You are advised to show your working.

(2)

OLevels.COM SCHOOL

8. Question 1 · part f · 2 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part f only.

Figure 1 shows the market for smartphones.

(g) Using the diagram below, draw the likely effect on the market for smartphones following an increase in productivity. Label the new curve, the new equilibrium price and the new equilibrium quantity.

(3)

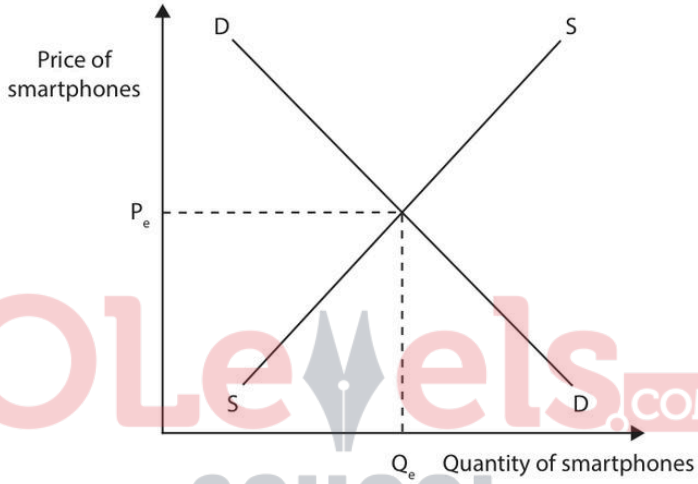


Figure 1

In 2023, the French Government increased the age at which people could receive a state pension to 64.

(h) Explain **one** possible effect of this change in the pension age on the supply of labour.

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

P 7 2 1 3 3 A 0 4 2 4

8. Question 1 · part f · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL